

Computational Docking and ADMET Analysis of Phytochemicals Against *Escherichia coli* DNA Gyrase B (PDB ID: 5L3J)

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Abstract

Antimicrobial resistance has become one of the most pressing global health concerns, demanding the discovery of new antibacterial agents. This study employed computational drug discovery approaches to evaluate the inhibitory potential of four naturally occurring phytochemicals **Berberine**, **Quercetin**, **Curcumin**, and **Resveratrol** against *Escherichia coli* DNA gyrase B (PDB ID: 5L3J), an essential bacterial enzyme involved in DNA replication. Molecular docking was performed using AutoDock Vina v1.2.7 to estimate ligand-binding affinities, while Discovery Studio Visualizer was used for protein–ligand interaction analysis. Pharmacokinetic and toxicity properties were predicted using pkCSM. **Berberine** demonstrated the strongest binding affinity (**-8.034 kcal/mol**), whereas **Quercetin** exhibited the most balanced pharmacokinetic and safety profile. These computational findings suggest that both compounds deserve further experimental investigation as potential antibacterial candidates.

Keywords: Molecular Docking, ADMET Prediction, DNA Gyrase B, AutoDock Vina, pkCSM, Phytochemicals, *Escherichia coli*

Introduction

The increasing prevalence of antimicrobial resistance has significantly reduced the effectiveness of conventional antibiotics and represents a major challenge to global healthcare. *Escherichia coli* is an important bacterial pathogen associated with urinary tract infections, gastrointestinal diseases, and bloodstream infections. DNA gyrase B is an essential bacterial enzyme responsible for DNA supercoiling and replication, making it an attractive molecular target for antibacterial drug discovery.

Natural phytochemicals have gained considerable attention because of their structural diversity and broad spectrum of biological activities. Computational

approaches such as molecular docking and ADMET prediction provide a rapid and cost-effective strategy for identifying promising lead compounds before experimental validation. This project integrates molecular docking with pharmacokinetic analysis to evaluate the antibacterial potential of selected phytochemicals against *E. coli* DNA gyrase B.

Materials and Methods

Protein Preparation

The crystal structure of *Escherichia coli* DNA gyrase B (PDB ID: 5L3J) was retrieved from the RCSB Protein Data Bank. The receptor structure was prepared by removing water molecules and unnecessary heteroatoms before conversion into the docking format.

Ligand Preparation

Four phytochemicals were selected for computational evaluation:

- Berberine
- Quercetin
- Curcumin
- Resveratrol

Ligand structures were prepared using standard molecular modeling procedures and converted into PDBQT format.

Molecular Docking

Molecular docking simulations were performed using AutoDock Vina v1.2.7.

Docking Parameters

- **Grid Center:** X = -11.95, Y = 20.49, Z = 23.40
- **Grid Size:** 20 × 20 × 20 Å
- **Exhaustiveness:** 8

The binding pose with the lowest binding energy was selected for each ligand.

Interaction Analysis

Protein–ligand interactions were analyzed using Discovery Studio Visualizer to identify hydrogen bonds, hydrophobic interactions, π -interactions, and van der Waals contacts.

ADMET Prediction

pkCSM were employed to predict physicochemical properties, absorption, distribution, metabolism, excretion, and toxicity of all selected phytochemicals.

Results

Molecular Docking

Compound	Binding Affinity (kcal/mol)	Rank
Berberine	-8.034	1
Quercetin	-7.635	2
Curcumin	-6.816	3
Resveratrol	-6.763	4

Among the investigated phytochemicals, Berberine demonstrated the strongest predicted binding affinity toward DNA gyrase B, suggesting the highest binding stability among the selected ligands. Quercetin also exhibited strong binding, while Curcumin and Resveratrol showed moderate docking performance.

ADMET Prediction

The computational ADMET analysis indicated that all four phytochemicals possess acceptable drug-like properties with distinct pharmacokinetic characteristics.

Berberine exhibited excellent intestinal absorption and the strongest docking affinity but showed predicted AMES toxicity and hepatotoxicity. **Quercetin** demonstrated a favorable balance between efficacy and safety with no predicted hepatotoxicity. **Curcumin** displayed acceptable pharmacokinetic behavior despite relatively low aqueous solubility, whereas **Resveratrol** exhibited high intestinal absorption but potential mutagenicity according to the AMES

prediction model. Overall, Quercetin provided the most balanced efficacy-to-safety profile among the evaluated compounds.

Discussion

The integration of molecular docking and ADMET prediction provides valuable insight into the therapeutic potential of natural compounds before laboratory experimentation. **Berberine** displayed the strongest predicted interaction with the DNA gyrase B active site, indicating high binding affinity. However, the predicted toxicity profile suggests that further safety evaluation is required.

Quercetin exhibited slightly lower binding affinity but demonstrated a more favorable pharmacokinetic and toxicity profile, making it a promising candidate for future experimental validation. **Curcumin** and **Resveratrol** also interacted with the target protein but produced comparatively weaker docking scores.

Overall, combining docking analysis with ADMET prediction allows more informed prioritization of lead compounds for antibacterial drug discovery.

Conclusion

This computational study successfully evaluated four phytochemicals against *Escherichia coli* DNA gyrase B using molecular docking and ADMET prediction. **Berberine** demonstrated the highest predicted binding affinity, while **Quercetin** exhibited the most balanced combination of binding affinity and predicted safety. These findings support the use of computational screening as an effective strategy for identifying potential antibacterial lead compounds and provide a foundation for future in vitro and in vivo investigations.

Software Used

- AutoDock Vina v1.2.7
- MGLTools 1.5.7
- Discovery Studio Visualizer
- PyMOL
- pkCSM

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GitHub: <https://github.com/rehanaly/Computational-docking-and-ADMET-analysis-of-phytochemicals-against-E.-coli-DNA-gyrase-B/tree/main/Data>